

 nova-3

Automatic Speech Recognition • Deepgram

@cf/deepgram/nova-3

Transcribe audio using Deepgram’s speech-to-text model

Model Info

Batch	Yes
Partner	Yes
Real-time	Yes
Unit Pricing	\$0.0052 per audio minute

Note

The [pricing of this model](#) is different based on transport. Transport-based pricing does not apply to all models.

- WebSocket: \$0.0092 per audio minute output (836.36 neurons per audio minute output)
- Regular HTTP: \$0.0052 per audio minute output (472.73 neurons per audio minute output)

Usage

▼ Worker

TypeScript

```
export default {
  async fetch(request, env, ctx):
    Promise<Response> {
      const URL =
        "https://URL_TO_MP3_FILE/audio.mp3";
      const mp3 = await fetch(URL);

      const resp = await
        env.AI.run("@cf/deepgram/nova-3", {
          "audio": {
            body: mp3.body,
            contentType: "audio/mpeg"
          },
          "detect_language": true
        }, {
          returnRawResponse: true
        });
      return resp;
    },
} satisfies ExportedHandler<Env>;
```

▼ curl

```
curl --request POST --url  
'https://api.cloudflare.com/client/v4/accounts/  
3?detect_language=true' --header 'Authorizat  
'Content-Type: audio/mpeg' --data-binary "@/
```

Parameters

* indicates a required field

Input

- audio **object** required
 - body **object** required
 - contentType **string** required
- custom_topic_mode **string**

Sets how the model will interpret strings submitted to the `custom_topic` param. When strict, the model will only return topics submitted using the `custom_topic` param. When extended, the model will return its own detected topics in addition to those submitted using the `custom_topic` param.

- `custom_topic` **string**

Custom topics you want the model to detect within your input audio or text if present Submit up to 100

- `custom_intent_mode` **string**

Sets how the model will interpret intents submitted to the `custom_intent` param. When strict, the model will only return intents submitted using the `custom_intent` param. When extended, the model will return its own detected intents in addition those submitted using the `custom_intents` param

- `custom_intent` **string**

Custom intents you want the model to detect within your input audio if present

- **detect_entities** **boolean**

Identifies and extracts key entities from content in submitted audio

- **detect_language** **boolean**

Identifies the dominant language spoken in submitted audio

- **diarize** **boolean**

Recognize speaker changes. Each word in the transcript will be assigned a speaker number starting at 0

- **dictation** **boolean**

Identify and extract key entities from content in submitted audio

- **encoding** **string**

Specify the expected encoding of your submitted audio

- **extra** **string**

Arbitrary key-value pairs that are attached to the API response for usage in downstream processing

- **filler_words** **boolean**

Filler Words can help transcribe interruptions in your audio, like 'uh' and 'um'

- **keyterm** **string**

Key term prompting can boost or suppress specialized terminology and brands.

- **keywords** **string**

Keywords can boost or suppress specialized terminology and brands.

- **language** **string**

The BCP-47 language tag that hints at the primary spoken language. Depending on the Model and API endpoint you choose only certain languages are available.

- **measurements** **boolean**

Spoken measurements will be converted to their corresponding abbreviations.

- **mip_opt_out** **boolean**

Opts out requests from the Deepgram Model

Improvement Program. Refer to our Docs for pricing impacts before setting this to true.

<https://dpgr.am/deepgram-mip>.

- `mode` **string**

Mode of operation for the model representing broad area of topic that will be talked about in the supplied audio

- `multichannel` **boolean**

Transcribe each audio channel independently.

- `numerals` **boolean**

Numerals converts numbers from written format to numerical format.

- `paragraphs` **boolean**

Splits audio into paragraphs to improve transcript readability.

- `profanity_filter` **boolean**

Profanity Filter looks for recognized profanity and converts it to the nearest recognized non-profane word or removes it from the transcript completely.

- **punctuate** **boolean**

Add punctuation and capitalization to the transcript.

- **redact** **string**

Redaction removes sensitive information from your transcripts.

- **replace** **string**

Search for terms or phrases in submitted audio and replaces them.

- **search** **string**

Search for terms or phrases in submitted audio.

- **sentiment** **boolean**

Recognizes the sentiment throughout a transcript or text.

- **smart_format** **boolean**

Apply formatting to transcript output. When set to true, additional formatting will be applied to transcripts to improve readability.

- **topics** **boolean**

Detect topics throughout a transcript or text.

- **utterances** **boolean**

Segments speech into meaningful semantic units.

- **utt_split** **number**

Seconds to wait before detecting a pause between words in submitted audio.

- **channels** **number**

The number of channels in the submitted audio

- **interim_results** **boolean**

Specifies whether the streaming endpoint should provide ongoing transcription updates as more audio is received. When set to true, the endpoint sends continuous updates, meaning transcription results may

evolve over time. Note: Supported only for websockets.

- **endpointing** **string**

Indicates how long model will wait to detect whether a speaker has finished speaking or pauses for a significant period of time. When set to a value, the streaming endpoint immediately finalizes the transcription for the processed time range and returns the transcript with a `speech_final` parameter set to true. Can also be set to false to disable endpointing

- **vad_events** **boolean**

Indicates that speech has started. You'll begin receiving Speech Started messages upon speech starting. Note: Supported only for websockets.

- **utterance_end_ms** **boolean**

Indicates how long model will wait to send an UtteranceEnd message after a word has been transcribed. Use with `interim_results`. Note: Supported only for websockets.

Output

- results **object**
 - channels **array**
 - items **object**
 - alternatives **array**
 - items **object**
 - confidence **number**
 - transcript **string**
 - words **array**
 - items **object**
 - confidence **number**
 - end **number**
 - start **number**

- word **string**
- summary **object**
 - result **string**
 - short **string**
- sentiments **object**
 - segments **array**
 - items **object**
 - text **string**
 - start_word **number**
 - end_word **number**
 - sentiment **string**
 - sentiment_score **number**
 - average **object**

- sentiment **string**
- sentiment_score **number**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{  
  "type": "object",  
  "properties": {  
    "audio": {  
      "type": "object",  
      "properties": {  
        "body": {  
          "type": "object"  
        },  
        "contentType": {  
          "type": "string"  
        }  
      },  
      "required": [  
        "body",  
        "contentType"  
      ]  
    },  
    "custom_topic_mode": {  
      "type": "string",  
      "enum": [  
        "extended",  
        "strict"  
      ],  
    }  
  }  
}
```



```
    "description": "Sets how the model  
will interpret strings submitted to the  
custom_topic param. When strict, the model  
will only return topics submitted using the  
custom_topic param. When extended, the model  
will return its own detected topics in  
addition to those submitted using the  
custom_topic param."
```

```
  },
```

```
    "custom_topic": {
```

```
      "type": "string",
```

```
      "description": "Custom topics you  
want the model to detect within your input  
audio or text if present Submit up to 100"
```

```
    },
```

```
    "custom_intent_mode": {
```

```
      "type": "string",
```

```
      "description": "Sets how the model  
will interpret intents submitted to the  
custom_intent param. When strict, the model  
will only return intents submitted using the  
custom_intent param. When extended, the model  
will return its own detected intents in  
addition those submitted using the  
custom_intents param",
```

```
    "enum": [  
      "extended",  
      "strict"  
    ],  
  },  
  "custom_intent": {  
    "type": "string",  
    "description": "Custom intents you  
want the model to detect within your input  
audio if present"  
  },  
  "detect_entities": {  
    "type": "boolean",  
    "description": "Identifies and  
extracts key entities from content in  
submitted audio"  
  },  
  "detect_language": {  
    "type": "boolean",  
    "description": "Identifies the  
dominant language spoken in submitted audio"  
  },  
  "diarize": {  
    "type": "boolean",
```

```
      "description": "Recognize speaker
changes. Each word in the transcript will be
assigned a speaker number starting at 0"
    },
    "dictation": {
      "type": "boolean",
      "description": "Identify and
extract key entities from content in submitted
audio"
    },
    "encoding": {
      "type": "string",
      "description": "Specify the
expected encoding of your submitted audio",
      "enum": [
        "linear16",
        "flac",
        "mulaw",
        "amr-nb",
        "amr-wb",
        "opus",
        "speex",
        "g729"
      ]
    },
  },
```

```
"extra": {  
  "type": "string",  
  "description": "Arbitrary key-value pairs that are attached to the API response for usage in downstream processing"  
},  
"filler_words": {  
  "type": "boolean",  
  "description": "Filler Words can help transcribe interruptions in your audio, like 'uh' and 'um'"  
},  
"keyterm": {  
  "type": "string",  
  "description": "Key term prompting can boost or suppress specialized terminology and brands."  
},  
"keywords": {  
  "type": "string",  
  "description": "Keywords can boost or suppress specialized terminology and brands."  
},  
"language": {
```

```
    "type": "string",
    "description": "The BCP-47
language tag that hints at the primary spoken
language. Depending on the Model and API
endpoint you choose only certain languages are
available."
  },
  "measurements": {
    "type": "boolean",
    "description": "Spoken
measurements will be converted to their
corresponding abbreviations."
  },
  "mip_opt_out": {
    "type": "boolean",
    "description": "Opts out requests
from the Deepgram Model Improvement Program.
Refer to our Docs for pricing impacts before
setting this to true.
https://dpgr.am/deepgram-mip."
  },
  "mode": {
    "type": "string",
    "description": "Mode of operation
for the model representing broad area of topic
```

that will be talked about in the supplied audio",

```
    "enum": [  
        "general",  
        "medical",  
        "finance"  
    ],  
},  
"multichannel": {  
    "type": "boolean",  
    "description": "Transcribe each  
audio channel independently."  
},  
"numerals": {  
    "type": "boolean",  
    "description": "Numerals converts  
numbers from written format to numerical  
format."  
},  
"paragraphs": {  
    "type": "boolean",  
    "description": "Splits audio into  
paragraphs to improve transcript readability."  
},  
"profanity_filter": {
```

```
    "type": "boolean",
    "description": "Profanity Filter
looks for recognized profanity and converts it
to the nearest recognized non-profane word or
removes it from the transcript completely."
  },
  "punctuate": {
    "type": "boolean",
    "description": "Add punctuation
and capitalization to the transcript."
  },
  "redact": {
    "type": "string",
    "description": "Redaction removes
sensitive information from your transcripts."
  },
  "replace": {
    "type": "string",
    "description": "Search for terms
or phrases in submitted audio and replaces
them."
  },
  "search": {
    "type": "string",
```

```
      "description": "Search for terms
or phrases in submitted audio."
    },
    "sentiment": {
      "type": "boolean",
      "description": "Recognizes the
sentiment throughout a transcript or text."
    },
    "smart_format": {
      "type": "boolean",
      "description": "Apply formatting
to transcript output. When set to true,
additional formatting will be applied to
transcripts to improve readability."
    },
    "topics": {
      "type": "boolean",
      "description": "Detect topics
throughout a transcript or text."
    },
    "utterances": {
      "type": "boolean",
      "description": "Segments speech
into meaningful semantic units."
    },
  },
```



```
"utt_split": {  
  "type": "number",  
  "description": "Seconds to wait  
before detecting a pause between words in  
submitted audio."  
},  
"channels": {  
  "type": "number",  
  "description": "The number of  
channels in the submitted audio"  
},  
"interim_results": {  
  "type": "boolean",  
  "description": "Specifies whether  
the streaming endpoint should provide ongoing  
transcription updates as more audio is  
received. When set to true, the endpoint sends  
continuous updates, meaning transcription  
results may evolve over time. Note: Supported  
only for websockets."  
},  
"endpointing": {  
  "type": "string",  
  "description": "Indicates how long  
model will wait to detect whether a speaker
```

has finished speaking or pauses for a significant period of time. When set to a value, the streaming endpoint immediately finalizes the transcription for the processed time range and returns the transcript with a `speech_final` parameter set to true. Can also be set to false to disable endpointing"

```
    },  
    "vad_events": {  
      "type": "boolean",  
      "description": "Indicates that  
speech has started. You'll begin receiving  
Speech Started messages upon speech starting.  
Note: Supported only for websockets."  
    },  
    "utterance_end_ms": {  
      "type": "boolean",  
      "description": "Indicates how long  
model will wait to send an UtteranceEnd  
message after a word has been transcribed. Use  
with interim_results. Note: Supported only for  
websockets."  
    }  
  },  
  "required": [
```

```
    "audio"  
  ]  
}
```

Was this helpful?



Previous

← Dashboard

Next

Workers Bindings →

Resources

[API](#)

[New to Cloudflare?](#)

[Directory](#)

[Sponsorships](#)

[Open Source](#)

Company

Support

[Help Center](#)

[System Status](#)

[Compliance](#)

[GDPR](#)

Tools

[cloudflare.com](#)

[Cloudflare Radar](#)

[Our team](#)

[Speed Test](#)

[Careers](#)

[Is BGP Safe Yet?](#)

[RPKI Toolkit](#)

[Certificate Transparency](#)

Community

 [X](#)

 [Discord](#)

 [YouTube](#)

 [GitHub](#)

© 2025 Cloudflare, Inc. • [Privacy Policy](#) • [Terms of Use](#)

• [Report Security Issues](#) • [Trademark](#) •  [Cookie Preferences](#)